

ENERGY PROPAGATION IN MICROTUBULES

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Abstract

Unlike the regular cytoskeletal structure in neurons, the cytoskeleton in most cells are highly dynamic to account for the different signaling pathways as well as structural needs. One would assume that the cytoskeletal network in cancer cells are highly dynamic and subject to change based on the cells' needs. Since there are heterogeneities in the cancer cells' energy requirements, the Warburg effect plays a crucial role in tumorigenesis by providing quick access to short bursts of energy to the different cells. However, in order to study this, we must first have a clear understanding of the energy translation within the cytoskeleton. The purpose of this paper is to develop a mathematical model to understand the energy propagation within the microtubules. This paper will study the quantum mechanical perspectives of energy propagation as well as the thermodynamics and electrodynamics aspect in the different parts of the microtubules (MT), especially, the C-tails and the nanopores at the end of the paper.

INTRODUCTION

Although cancer cells are always destructive to an organism they reside in, the cells themselves are very efficient and self-sufficient. They eventually adapt and acquire the hallmarks of cancer that allows them to respond productively to a negative action from the host organism. When patients are diagnosed with a late-stage cancer and are unresponsive to any form of cancer therapy or treatment, the diagnosis often refers to metastatic cancer. The unresponsiveness likely results from the adaptations the cancer cells acquire that allows them to resist the treatment. One would assume a wide range of signaling pathways are involved that differ from patient to patient. This level of complexity is often analogous to the complexity of the interconnected web-like structures that form the cytoskeleton. This cytoskeleton forms the backbone of the cytoplasm and is essentially responsible, in one way or the other, for most of the activities in the cell. Due to the heterogeneity in the phenotypes of the cancer cells, there is also heterogeneity in the architectural network of the cytoskeleton and, in turn, the microtubules' dynamic structure.

Elongation of microtubule at the plus and minus ends of the protofilament occurs via addition of tubulin dimers. As GTP-tubulin is added to the protofilament the GTP in its β monomer gets hydrolysed immediately. This hydrolysis, summarised below, releases about 0.42eV⁴. However, only 0.17 eV⁴ of this energy is available as Gibbs free energy while the rest of the energy is stored in the conformation of the tubulin dimer⁴. This Gibbs free energy may be coupled to vibrational oscillations within the protofilament⁵. This paper will discuss in detail how these oscillations might couple to the dynamic nature of cell signalling pathways, particularly in pathways concerning tumorigenesis.



Each tubulin subunit measures 8nm×4.5nm×4.5nm and consists of α and β tubulins monomer, each measuring at 4.5nm×5.5nm and 55,000 Da of molecular weight. Each monomer has a C-terminal tail that is 4.5nm long and 1nm in diameter¹. A single microtubule (MT) consists of 13 protofilaments so that the total diameter of an MT is 25nm with an inner diameter of 15nm. In between, the protofilaments there are nanopores – subject to study later in the paper- that permeable to small ions. This paper first looks at the electronic conduction within the MT itself,

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then at the ionic wave conduction outside the MT along the protofilament axis, and then finally analyses this ionic wave conduction with respect to the nanopores and the C-tails.

1. MODELLING PROTOFILAMENT CHAIN

Tuszynski et al. 2005¹ found out that there are hydrophobic pockets beside each monomer. Along with the conclusion from their simulations that the outer surfaces of the monomers are mostly hydrophilic, these hydrophobic pockets may give rise to double quantum well structure, and indeed their TINKER simulations¹ supported the idea of quantum double wells. These quantum wells may facilitate electron tunnelling within the dimer and across dimers.

With a semiclassical view on the movement of electrons along the protofilament, we assume that there are two conduction electrons for each dimer, with one electron attributed to each monomer³. As a result, the conduction density for the dimer can be given by:

$$n_{density} = \frac{2}{\pi r^2 h} = 1 \cdot 27 \times 10^{25} \text{m}^{-3}$$

which is higher than the typical conduction density for a semiconductor but quite lower than a conductor. However, these electrons are likely to have very low mobilities unless they are excited from their ground state to conduction bands. Therefore, some electrons can be in the ground state while some in the conduction bands. The electron conduction density, n_B can then be found with respect to the Boltzmann factor:

$$n_B = n_{density} e^{-\frac{\Delta}{k_B T}}$$

where, $e^{-\frac{\Delta}{k_B T}}$ is the Boltzmann factor and n_B is the electron conduction density with respect to this factor and Δ has a value of 0.4eV for exciting an electron from a ground state to a conduction band

Now, we assume that the electrical mobilities for the dimers would be similar to semiconductors and be around, $u_0 = 0.13 \text{m}^2 \text{V}^{-1} \text{s}^{-1}$. The electrical conductivity, ζ can be found by the following formula:

$$\zeta = n_B e u_0$$

where, e is the charge of an electron. This results in an electron conductivity of $0.08 \Omega^{-1} \text{m}^{-1}$ (approx..) at physiological temperatures. However, this value may increase to about $2 \times 10^5 \Omega^{-1} \text{m}^{-1}$ as electrons hybridize and the gap between the ground state and the conduction band becomes negligible. In order to understand the possible semiconductor properties of

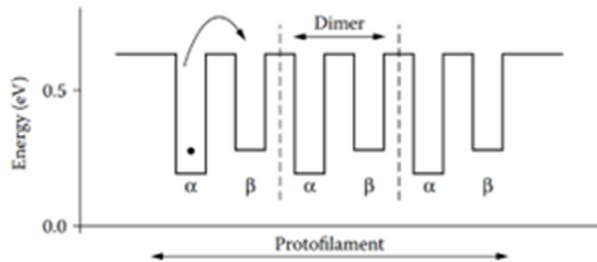


Figure 1. One-dimensional quantum double well representation of protofilament with 25% difference in well depth³

microtubule filaments, one must first understand the expected charge movements within tubulin dimers and between dimers via electron tunneling in a quantum double-well scenario where each monomer is viewed a quantum well. The alpha-monomer is 0.17eV deeper than the beta monomer. This distinctiveness arises from the GTP hydrolysis in β monomer as mentioned. Binding energy between the dimers must be higher than the 0.17eV released to the

microtubules to maintain the lattice structure of microtubules³. Hopping between the quantum wells (figure 1) must have energies comparable to these binding energies. Intra-dimer hopping and inter-dimer hopping potentials, V_0 has been estimated to be 0.4eV and 1.0 eV respectively³. The time-independent Schrodinger's equation for a quantum well with the piece-wise constant potential^{figure 1} would given by:

$$\hat{H}\psi(x) = \frac{-\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \hat{V}\psi(x) = E\psi(x) \quad (1.1)$$

with the oscillatory solution for $E > V$:

$$\psi = A \sin kx + B \cos kx$$

where

$$k^2 = \frac{2m(E - V)}{\hbar^2}$$

and for $E < V$, the solution is damped:

$$\psi(x) = Ce^{\kappa x} + De^{-\kappa x}$$

and,

$$\kappa^2 = \frac{2m(V - E)}{\hbar^2}$$

Since the number of quantum double-wells are increased (as dimer number for the microtubules increases), one must resort to second quantization to determine the electron eigenstates of the many-body system. Such a system can be evaluated from the Hubbard Hamiltonian model – a simple second quantized model that would determine the essential features in a system of interacting particle⁸ – like our system with interacting electrons represented as fundamental excitations. The second quantized Hamiltonian for the Hubbard model is given by:

$$\hat{H} = \sum_{i,\sigma} \epsilon_i \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} - \sum_{i \neq j, \sigma} t_{ij} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \sum_i U_i \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} \quad (1.2)$$

where ϵ_i represents the potential energy at each site i and t_{ij} represents the kinetic energy to move an electron from site i to site j . σ is the spin of the electron and $\sigma = -\frac{1}{2}, 0, +\frac{1}{2}$. i ranges from 1 to $2n$ to represent the two electron sites on n dimers, assuming $i \neq j$. The creation and annihilation operators for the electrons are given by fermion operators $\hat{c}_{i\sigma}^\dagger$ and $\hat{c}_{i\sigma}$ respectively. The number of electronic states can be determined from the number of binding sites $2n$ and the number of electrons, e^- to obtain

$$\binom{2n}{e^-} = \frac{2n!}{e^-! (2n - e^-)!}$$

number of electronic states. With a two-electron system for a single dimer the possible wavefunctions $|\psi\rangle$ can be written in the basis $\{|\alpha\alpha\rangle, |\alpha\beta\rangle, |\beta\alpha\rangle, |\beta\beta\rangle\}$, with the basis states being a combination of the localized single-electron states of $|\alpha\rangle$ and $|\beta\rangle$ - one electron either in the α well or the β well in the ground state. For such a system, let us assume the α site as 1 ($i=1$) and

the β site as $2(i=2)$. In the absence of the interaction energy, the Hamiltonian in equation 1.2 can be diagonalized by the following unitary transformations³

$$\begin{aligned}\alpha_{1'\sigma}^\dagger &= \frac{1}{\sqrt{2}}(\hat{c}_{1\sigma}^\dagger + \hat{c}_{2\sigma}^\dagger) \\ \alpha_{2'\sigma}^\dagger &= \frac{1}{\sqrt{2}}(\hat{c}_{1\sigma}^\dagger - \hat{c}_{2\sigma}^\dagger)\end{aligned}\quad (1.3)$$

where the $1'$ and $2'$ are new sites created for the new operators $\alpha_{1'\sigma}^\dagger$ and $\alpha_{2'\sigma}^\dagger$, to differentiate from the physical sites: 1 and 2. Furthermore, these new operators can be shown to obey anti-commutation relation for fermions. Substituting these new operators onto Equation (1.2) generates³:

$$\hat{H} = \sum_{\sigma} (\epsilon + t) \alpha_{1'\sigma}^\dagger \alpha_{1'\sigma} + \sum_{\sigma} (\epsilon - t) \alpha_{2'\sigma}^\dagger \alpha_{2'\sigma} + \frac{1}{2} \sum_{\substack{i,j,l,m \\ \sigma \neq \sigma'}} V_{ijlm} \alpha_{i\sigma}^\dagger \alpha_{j\sigma}^\dagger \alpha_{l\sigma'} \alpha_{m\sigma'} \quad (1.4)$$

where the site energies are made equivalent, i.e., $\epsilon_1 = \epsilon_2 = \epsilon$.

To evaluate the interaction energy for Equation (1.1), the form of V_{ijlm} is solved. When the spins $\sigma = \sigma'$, then V_{ijlm} is found to be zero. Therefore, for Equation (1.3) we are only concerned with spin states where $\sigma \neq \sigma'$. This signifies that the coulomb interaction preserves the electron spin. As a result, V_{ijlm} can be found for two scenarios:

Even combination of indices for V_{ijlm} , e.g. $V_{1122} = \frac{1}{4}(U_1 + U_2)$

Odd combination of indices for V_{ijlm} , e.g., $V_{1222} = \frac{1}{4}(U_1 - U_2)$

Considering the 2 electrons, each having a spin of $+\frac{1}{2}, 0, -\frac{1}{2}$, there are 6 possible combinations of electron states, which are used to study Equation (1.4) modified with the new operators:

$$\begin{aligned}|1\rangle &= \alpha_{1\uparrow}^\dagger \alpha_{1\downarrow}^\dagger |\Omega\rangle & |4\rangle &= \alpha_{1\downarrow}^\dagger \alpha_{2\uparrow}^\dagger |\Omega\rangle \\ |2\rangle &= \alpha_{1\uparrow}^\dagger \alpha_{2\uparrow}^\dagger |\Omega\rangle & |5\rangle &= \alpha_{1\downarrow}^\dagger \alpha_{2\downarrow}^\dagger |\Omega\rangle \\ |3\rangle &= \alpha_{1\uparrow}^\dagger \alpha_{2\downarrow}^\dagger |\Omega\rangle & |6\rangle &= \alpha_{2\uparrow}^\dagger \alpha_{2\downarrow}^\dagger |\Omega\rangle\end{aligned}\quad (1.5)$$

where $|\Omega\rangle$ is the empty ground state. Acting on these basis states, the following Hamiltonian matrix was obtained

$$\begin{pmatrix} 2(\epsilon + t) + \frac{U_1 + U_2}{4} & 0 & \frac{U_1 - U_2}{4} & \frac{U_2 - U_1}{4} & 0 & \frac{U_1 + U_2}{4} \\ 0 & 2\epsilon & 0 & 0 & 0 & 0 \\ \frac{U_1 - U_2}{4} & 0 & 2\epsilon + \frac{U_1 + U_2}{4} & -\frac{(U_1 + U_2)}{4} & 0 & \frac{U_1 - U_2}{4} \\ \frac{U_2 - U_1}{4} & 0 & -\frac{(U_1 + U_2)}{4} & 2\epsilon + \frac{U_1 + U_2}{4} & 0 & \frac{U_2 - U_1}{4} \\ 0 & 0 & 0 & 0 & 2\epsilon & 0 \\ \frac{U_1 + U_2}{4} & 0 & \frac{U_1 - U_2}{4} & \frac{U_2 - U_1}{4} & 0 & 2(\epsilon - t) + \frac{U_1 + U_2}{4} \end{pmatrix}$$

This Hamiltonian is diagonalized to find the energy eigenstates. In the absence of the Coulomb interaction, the two electrons are found either in the $1'$ site with energies $2(\epsilon + t)$ or the $2'$ site with energies $2(\epsilon - t)$ since opposite spin electrons have zero interaction energy. However, in the presence of Coulomb interaction energy(repulsion), the energy gap between the ground state and

the conductive band is reduced. From diagonalization of the above matrix there are 6 possible eigenstates, which is evident from the 3 singlet states of our two-electron system (figure 5). The singlet state is defined by two electrons with opposite spin on the same binding site, or the state consists of an antisymmetric combination of up and down electrons on neighboring binding sites.

For the double quantum well, there are two kinds of hopping parameters for the system: t_{intra} and t_{inter} for electrons hopping within dimers and hopping between dimers respectively. However, the hopping parameter is set at $t = 0.4\text{eV}$ from the 1D Schrodinger equation (Equation 1)³. Since the intradimer hopping parameter, t_{intra} , is the largest, the system keeps the two electrons of the same spin within the same dimer in the ground state. As a result, only singlet with anti-symmetric combination is allowed when $t_{inter} < t_{intra}$. However, introduction of a kinetic term allows the introduction of paired electrons in the same site as the Coulomb repulsion is competing with the kinetic term. This is further explained in the t_{inter}/t_{intra} ratios which determines the energy gap between the ground state and the 1st excited state. When the t_{inter} is small relative to t_{intra} , this energy gap is quite large preventing paired electrons in the same site. However, as t_{inter} increases and approaches near the value of t_{intra} , the energy gap between bound state and the 1st excited state is minimized, allowing singlet states 1 and 3 illustrated in figure 2.

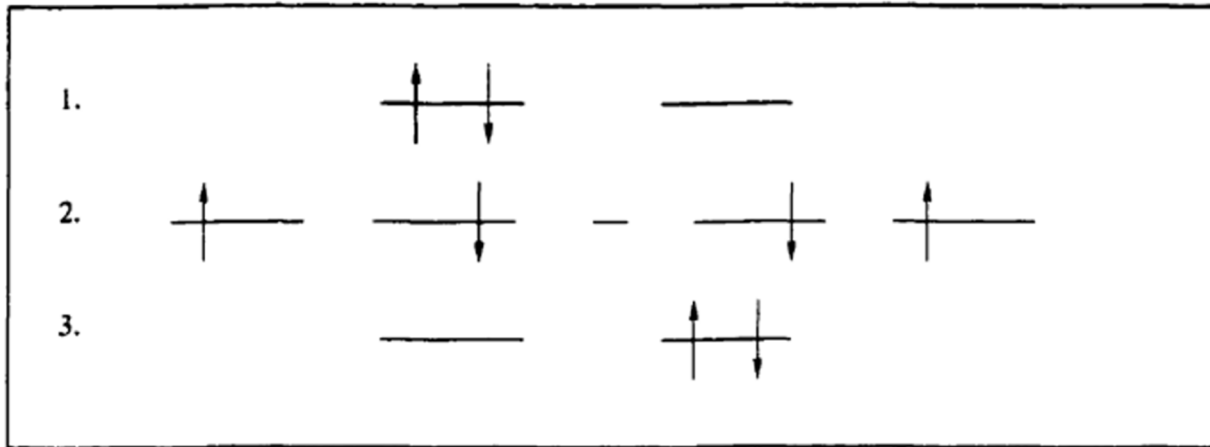


Figure2: The three singlet states for two electrons on the two sites of the tubulin dimer consist of paired electrons spins on the α site, paired electrons on the β site or the anti-symmetric combination of paired electrons on separate sites

2. CONDUCTION IN HUBBARD MODEL

Now that we have described the protofilament chain in terms of a series of quantum double wells in the Hubbard model, let us move onto describing conduction within the Hubbard model. Since ferroelectric alignment is usually found in microtubules(12), it would be useful to consider the effect of an external field to its conduction. We will work in the Coulomb gauge and introduce a vector, $A(t)$ that couples the electric field, $Ee^{-i\omega t}$ to our Hubbard Hamiltonian (Equation 1.2):

$$A(t) = \frac{cE(t)}{-i\omega} e^{-i\omega t} \quad (2.1)$$

where c is the speed of light and t is time. Similar representations are used for other quantities which will be referred to accordingly. A canonical transformation exists for a tight binding system where coupling term above is used to transform our annihilation operator, $\hat{c}$ such that:

$$\hat{c}'_i = \hat{c}_i e^{-ieA(t)x_i/c} \quad (2.2)$$

We will only consider the kinetic term of the Hubbard Hamilton for our analysis, where the kinetic term is referred to as $\hat{H}_k$. Transforming with our vector potential results in the following:

$$\hat{H}'_k = -t \sum_{i\sigma} [\hat{c}_{i\sigma}^\dagger \hat{c}_{i+1\sigma} e^{-ieA(t)/c} + \hat{c}_{i\sigma}^\dagger \hat{c}_{i-1\sigma} e^{ieA(t)/c}] \quad (2.3)$$

where t is the hopping parameter and canonical transformation reduces our vector potential to a static vector potential. Expanding to second order in A gives us:

$$\hat{H}_A = \frac{A}{c} \hat{j} - 2 \frac{A^2}{c^2} a^2 \hat{H}_k \quad (2.4)$$

where c is the speed of light and the current operator $\hat{j} = ieat \sum_{i\sigma} (\hat{c}_{i\sigma}^\dagger \hat{c}_{i+1\sigma} - \hat{c}_{i\sigma}^\dagger \hat{c}_{i-1\sigma})$

There is experimental evidence that single MTs can amplify the electrical signals in a system²⁰, which is why, we will focus our attention into how electrical fields can affect the conduction model put forward in this paper. Suppose an external field is applied to our MT system, then the induced current is given by:

$$J_\alpha(r, t) = \sum_\beta \sigma'_{\alpha\beta}(q, \omega) \Xi_\beta^{ext} e^{iq \cdot r - i\omega t} \quad (2.5)$$

where $\Xi_\beta^{ext} e^{iq \cdot r - i\omega t}$ is the external field, $\sigma'_{\alpha\beta}$ is the parameter that relates the direction of the external field to the current density, and α and β are Cartesian coordinate directions. However, we must account the effect of the internal electric field our MT system (explained in a later section). Then, we introduce the parameter $\sigma_{\alpha\beta}$ that relates all the electric fields in our system to its current densities giving the following form of Equation (2.5):

$$J_\alpha(r, t) = \sum_\beta \sigma_{\alpha\beta}(q, \omega) E_\beta e^{iq \cdot r - i\omega t} \quad (2.6)$$

The evolution of the current operator, $\hat{j}$, mentioned previously, with time is given by:

$$\hat{j}(q, t) = e^{i\hat{H}t} \hat{j}(q) e^{-i\hat{H}t} \quad (2.7)$$

The Kubo formula for electrical conductivity employs this current operator to evaluate the parameter, $\sigma_{\alpha\beta}$ such that²⁶:

$$\sigma_{\alpha\beta} = \frac{1}{\omega} \int_0^\infty dt e^{i\omega t} \langle \Omega | \hat{j}_\alpha^\dagger(q, t) \hat{j}_\beta(q, 0) - \hat{j}_\beta(q, 0) \hat{j}_\alpha^\dagger(q, t) | \Omega \rangle + i \frac{n_0 e^2}{m\omega} \delta_{\alpha\beta} \quad (2.8)$$

However, we will focus on the first term of the equation which is known as the incoherent contribution to the conductivity. Using the unity $\sum_n \langle n | n \rangle = 1$, we obtain the first term as:

$$\begin{aligned} & \sum_n \int_0^\infty dt e^{i\omega t} e^{i(E_\eta - E_n)t} \langle \Omega | \hat{j}_\alpha^\dagger(q, 0) | n \rangle \langle n | \hat{j}_\beta(q, 0) | \Omega \rangle \\ &= \sum_n \int_0^\infty dt e^{i\omega t} e^{i(E_\eta - E_n)t} |\langle n | \hat{j}_\alpha(q, 0) | \Omega \rangle|^2; \quad \alpha = \beta \end{aligned} \quad (2.9)$$

Where the eigenstates of our Hamiltonian, H acts on these states and produces the eigen-energies, E_n . We then, normalize Equation (2.9) using the partition function, Z on each of the energy eigenstates

$Z = \sum_m e^{-\frac{E_m}{k_B T}}$, so that Equation (2.9) becomes:

$$\left[\sum_{m,n} \int_0^\infty dt e^{-\frac{E_m}{k_B T}} e^{i\omega t} e^{i(E_m - E_n)t} |\langle n | \hat{j}_\alpha(q, 0) | \Omega \rangle|^2 \right] / Z \quad (2.10)$$

We can, now, use this expression to evaluate $\sigma_{\alpha\beta}$ from Equation (2.8). However, since we are only interested when the induced current is in the same direction as the electric field, $\sigma_{\alpha\alpha}$ is adopted instead to have the following equation:

$$\begin{aligned} \sigma_{\alpha\alpha} &= \sum_{m,n} \frac{1}{\omega} \int_0^\infty dt e^{i\omega t} e^{-\frac{E_m}{k_B T}} [e^{i(E_m - E_n)t} |\langle n | \hat{j}_\alpha(q, 0) | m \rangle|^2 - e^{i(E_m - E_n)t} |\langle m | \hat{j}_\alpha(q, 0) | n \rangle|^2] / Z \\ \sigma_{\alpha\alpha} &= \sum_{m,n} \frac{1}{\omega} \int_0^\infty dt e^{i\omega t} (e^{-\frac{E_m}{k_B T}} - e^{-\frac{E_n}{k_B T}}) [e^{i(E_m - E_n)t} |\langle n | \hat{j}_\alpha(q, 0) | m \rangle|^2] / Z \end{aligned} \quad (2.11)$$

3. DAVYDOV'S SOLITON IN DIMERS

The energy that is released from GTP hydrolysis is either stored in the closed conformation of the tubulin dimer or available in the form of Gibb's free energy. This free energy amounts to about 0.17eV(5), which is equivalent to the quantum energy of 0.2eV found in the vibrations of the peptide bond of the alpha-helix in proteins(9), so these vibrations are ideal to store or transport the free energy from GTP hydrolysis.

However, in order to propagate, these packets of energy in the form of vibrations must be non-dispersive. A simple linear wave equation of the form⁹

$$\frac{\delta y}{\delta t} + \frac{\delta^3 y}{\delta t^3} = 0 \quad (3.1)$$

can be used to describe the shape of the wave packet but due to the linearity of Equation (3.1), the wave packet will be dispersed and become nonlocalized. As a result, this wave packet will redistribute itself as the system containing this wave-packet reaches equilibrium.

Introducing a non-linear term such as $y \frac{\delta y}{\delta x}$ into Equation (3.1) can prevent the dispersion. As such, this equation then becomes

$$\frac{\delta y}{\delta t} + y \frac{\delta y}{\delta x} + \frac{\delta^3 y}{\delta t^3} = 0 \quad (3.2)$$

with the solution:

$$y(x, t) = 3v \operatorname{sech}^2[\sqrt{v}(x - vt)/2]$$

where, v is the propagation velocity. This solution is non-dispersive and localized. At time = 0, a pulse is formed for the wave-packet with the shape:

$$y(x, 0) = 3v \operatorname{sech}^2[\sqrt{v}(x)/2]$$

and does not disperse with time⁹. Such a wave-packet is termed a soliton and exist at non-equilibrium states in a system.

As mentioned previously in this section, vibrational quantum in the peptide bonds of alpha-helix hold about 0.2eV of energy. Furthermore, the peptide bonds have a resonance structure as illustrated in figure 2 that can result in dipole moments. In turn, dipole-dipole interactions with other proteins or molecules may result in the dispersion of the wave-packet. However, this dispersion must occur at a picosecond time frame, which is too short for biological significance⁹.

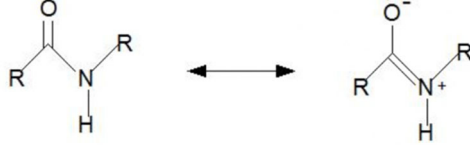


Figure 3. Resonance structure in the peptide bond. R can represent any combination of atoms to form the rest of the amino acid chain

The probability amplitude of finding our wave-packet in the form of vibrational quantum can be given by $\phi(n_\alpha\alpha)$ where n_α is the index for the number of helix turns and α (1, 2, or 3) counts the peptide groups in each turn. The probability of finding the vibrational quantum at $n_\alpha\alpha$ is then given as $|\phi(n_\alpha\alpha)|^2$. In the absence of dipole-dipole interactions, the solution to $\phi(n_\alpha\alpha)$ can be found from Schrodinger's time dependent equation:

$$i\hbar\phi(n_\alpha\alpha, t) = E_o\phi(n_\alpha\alpha) \quad (3.3)$$

where E_o is the energy of the vibrational quantum and such an equation has a solution:

$$\phi(n_\alpha) \propto e^{\frac{-iE_o t}{\hbar}}$$

In the presence of dipole-dipole interactions Equation(3.3) becomes⁹

$$i\hbar\phi(n_\alpha\alpha, t) = E_o\phi(n_\alpha\alpha) - J[\phi_{n_\alpha+1,\alpha} + \phi_{n_\alpha-1,\alpha}] + L[\phi_{n_\alpha,\alpha+1} + \phi_{n_\alpha,\alpha-1}] \quad (3.4)$$

where J is the strength of the longitudinal dipole-dipole interactions and L is the strength of the lateral dipole-dipole interactions. This equation is highly dispersive.

Davydov and Kishukha(11) proposed a mechanism for preventing the dispersion in Equation (3.4) via a non-linear effect that involves interaction with longitudinal sound waves or stretching of the hydrogen bonds in the alpha helix. In the absence of the non-linear effect, the longitudinal sound waves are given by

$$mz(n_\alpha, tt) - k[z(n_\alpha + 1, \alpha) - 2z(n_\alpha, \alpha) + z(n_\alpha - 1, \alpha)] = 0 \quad (3.5)$$

where z is the longitudinal displacement of the peptide group at n_α , m is the mass of the peptide group and the residue and k is the spring constant of a hydrogen bond.

The non-linear effect involved stretching the hydrogen-bond by the vibrational quantum. In other words, energy from the vibrations C=O groups in the peptide bonds are being used to extend the corresponding hydrogen bonds as depicted in figure 3. This effect can be expressed as the non-linear parameter, χ

$$\chi = \frac{dE_o}{dR} \quad (3.6)$$

where E_0 is the energy of the vibrational quantum in joules and R is the length of the hydrogen bond in meters. As a result, χ is the force in newtons acting on the hydrogen bond. Amending Equations (3.4) and (3.5) with our parameter, χ results in the following expressions:

$$i\hbar\phi_{n_\alpha\alpha,t} = [E_0 + \chi(z_{n_{\alpha+1},\alpha} - z_{n_\alpha,\alpha})]\phi_{n_\alpha,\alpha} - J[\phi_{n_{\alpha+1},\alpha} + \phi_{n_{\alpha-1},\alpha}] + L[\phi_{n_{\alpha,\alpha+1}} + \phi_{n_{\alpha,\alpha-1}}] \quad (3.7a)$$

$$mz_{n_\alpha,tt} - k[z_{n_{\alpha+1},\alpha} - 2z_{n_\alpha,\alpha} + z_{n_{\alpha-1},\alpha}] = -\chi[|\phi(n_\alpha)|^2 - |\phi(n-1, \alpha)|^2] \quad (3.7b)$$

where the subscripts $n_\alpha\alpha$, $n_\alpha + 1 \alpha$ and $n_\alpha - 1 \alpha$, denote the appropriate variable with the location of the peptide groups; the subscripts t and tt denote the first order and second order derivative with respect to time, k denotes the spring constant and, as before, z denotes the displacement of the longitudinal sound wave.

Now, let us talk about how the free energy from GTP hydrolysis mentioned before acts like a soliton. Once this energy is available, it may be translated into the vibrational quantum of the amide or carbonyl group in the peptide bond. When the vibrational quantum is localized in one of the peptide groups right-hand side quantity of Equation (3.7) will be a non-zero value and act as longitudinal sound waves. Then these longitudinal waves act via $\chi(z_{n_{\alpha+1},\alpha} - z_{n_\alpha,\alpha})$ to trap the localized vibrational quantum energy in a quantum well to create a soliton. The soliton is then prevented from dissipating from the dipole-dipole interactions presented in the last two terms of Equation (3.7b). However, the vibrational quantum, E_0 must reach a particular threshold, for example, E_{min} , so that it can act through χ on the right-hand side of Equation 7b and create a longitudinal sound wave with a threshold amplitude.

Aromatic amino acids including tryptophan, tyrosine, phenylalanine and histidine are ideal for electron tunneling due to the resonance in the benzene/non-benzene rings in these amino acids. However, tryptophan has the greatest resonance due to its extra indole ring, so it has the highest fluorescence among the four amino acids. As illustrated in figure 3, most of the alpha helix secondary structures in the tubulin dimer are found on its outer surface. Tryptophan residues is shown to be found mostly in the alpha helix with some residue in the beta sheet. Foerster distance limits electron excitation travel to under 10Å in an inert medium like ions, but within a protein, electrons can move via bond hopping (electron hopping) over more than 30Å¹² Figure 4A shows that there are distances of 30Å and above between some the tryptophan residues in the dimer. The figure also signifies that there are multiple pathways for electron hopping. Figure 4B showed that there are more histidine residues in the dimer than tryptophan, which also increases the number of routes for electron hopping since the hopping may not be restricted to a single aromatic residue but can occur within multiple residues. These aromatic residues can facilitate the conduction of excited electrons(created from GTP hydrolysis) via electron hopping and when these electrons reach peptide bonds, the vibrational quantum, that exists as a result, can remain self-trapped in soliton states via the non-linear parameter, χ introduced in Equation (3.6). Furthermore, ordered water molecules around the dimers can screen the dimers from further counter-ions and facilitate in maintaining these solitons.

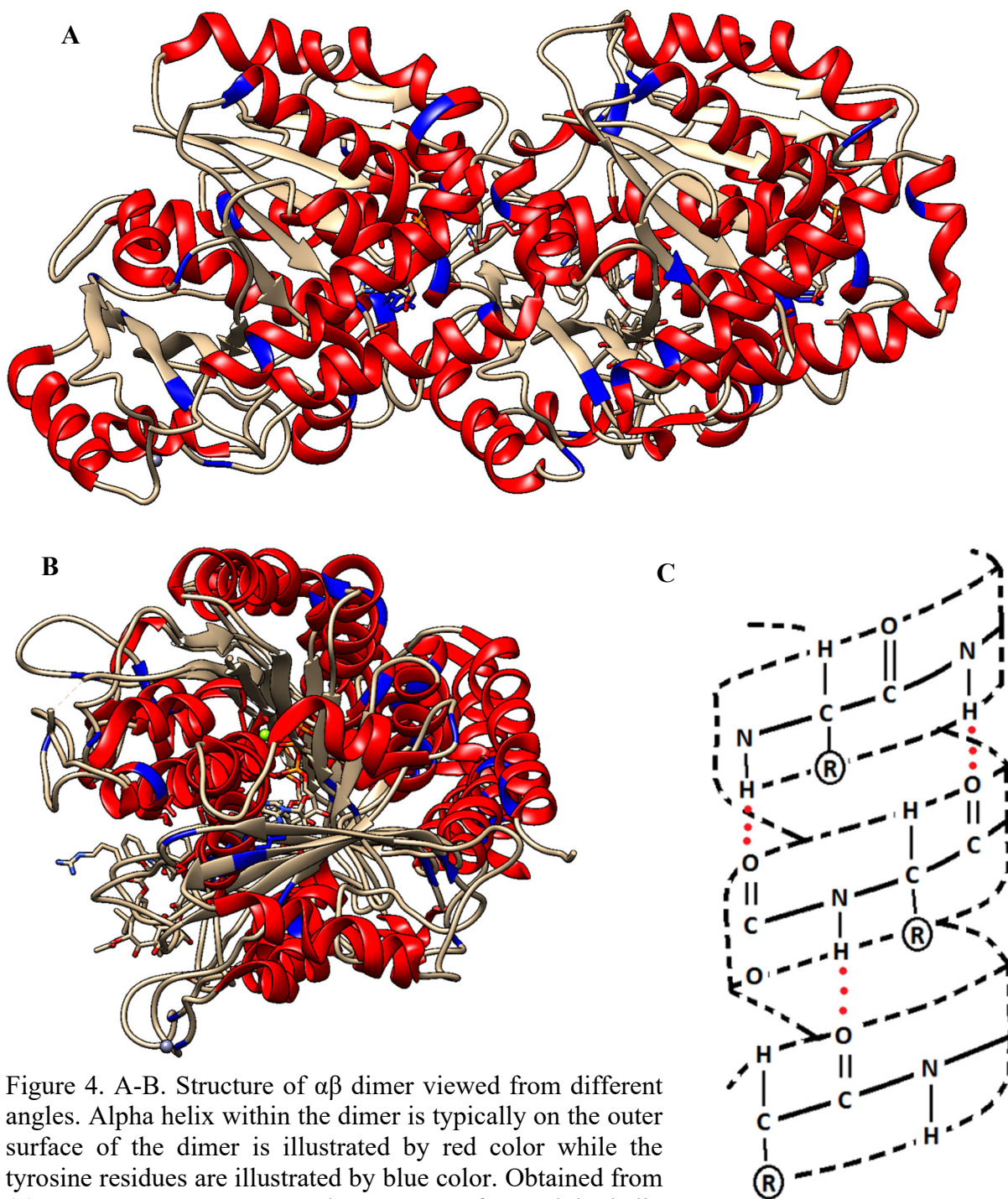


Figure 4. A-B. Structure of $\alpha\beta$ dimer viewed from different angles. Alpha helix within the dimer is typically on the outer surface of the dimer is illustrated by red color while the tyrosine residues are illustrated by blue color. Obtained from (7), PDB: J1FF C. General structure of an alpha-helix secondary structure. The red dots indicate the hydrogen bonds within the helix between an amide group and a carbonyl group.

4. FROHLICH'S COHERENCE

Now that we have established how solitons might exist within the dimer, now elucidate how different solitons might relate to one another. This is where work done by H. Frohlich¹⁴ in the 1970s comes into play.

Any charged materials carrying dipoles are capable of electric vibrations if the system allows for elastic deformity⁵. In our model, electric vibrations can therefore be maintained in the peptide bonds as the resonance would allow dipole creation, and the hydrogen bond stretching would allow elastic deformity. Coherence can then exist within the system if a certain threshold for energy supply rate is crossed. However, this energy threshold might not be equivalent to the threshold for soliton, (E_{min}).

In a Bose-Einstein condensation, as the temperature of a Bose gas is decreased below a certain threshold, the particles in the gas condense into a single quantum state. In a similar situation, keeping temperature constant, number of particles in the form of quanta, $\hbar\omega_k$ are added to the system and redistributed to other degrees of freedom in the system until the system reaches a static state where all the electric modes in the system have a higher energy than the thermal equilibrium. This difference in energy in the system is channeled into a single mode with the lowest frequency. The MT network could, in fact, carry electrical oscillations across the whole network within the cell¹³.

Consider numbers of Z peptide bonds vibrational oscillators in a system that is acting as a heat bath, capable of electric dipole oscillations because of their resonance. Each of these oscillators has a minimum frequency, ω_0 . Long-range Coulomb interactions between our oscillators would likely results in Z longitudinal electric modes in a frequency range $\omega_1 \leq \omega \leq \omega_2$. These electrical modes will likely not interact with one another but instead interact with the surrounding (acting as heat baths) with temperature, T .

Let n_k be the number of quanta, $\hbar\omega_k$ in modes ω_k . Then, discrete quanta can be absorbed or emitted from our heat-bath. In a first-order process, the rate of change of n_k is given by⁶:

$$\dot{n}_{k,1} = -\varphi[n_k e^{\beta\hbar\omega_k} - (n_{k+1})] \quad (4.1)$$

where $\beta=1/kT$ and φ may depend on T .

For a second-order process, the rate of change of n_k is given by⁶:

$$\dot{n}_{k,2} = -\chi_t \sum_j [n_k(1 + n_j)e^{\beta\hbar\omega_k} - n_j(1 + n_k)e^{\beta\hbar\omega_j}] \quad (4.2)$$

where χ_t is the transition probability that depends on i and j but shall be neglected in the calculations. Equations (4.1) and (4.2) require that, in the absence of energy supplied to the system, thermal equilibrium, defined by $\dot{n}_{k,1} = 0$ and $\dot{n}_{k,2} = 0$, follows the Planck distribution, n_{kT} :

$$n_{kT} = \frac{1}{e^{\beta\hbar\omega_k} - 1} \quad (4.3)$$

Consider that energy is supplied to our system at a rate, s_k . For reasons discussed later, let us assume $s_k = 0.17eV$ for per GTP molecule hydrolyzed. Then, the total rate change of n_k is given by:

$$\dot{n}_k = s_k - \varphi[n_k e^{\beta\hbar\omega_k} - (n_{k+1})] - \chi_t \sum_j [n_k(1 + n_j)e^{\beta\hbar\omega_k} - n_j(1 + n_k)e^{\beta\hbar\omega_j}] \quad (4.4)$$

For a stationary system we would assume that $\dot{n}_k = 0$. Now we define the total energy, S supplied to a particular mode, k with frequency ω_k as:

$$S = \sum s_k$$

Introducing this onto Equation (4.4) and assuming that $\dot{n}_{k,2} = 0$ yields:

$$S = \varphi \sum_k [n_k e^{\beta \hbar \omega_k} - (n_k + 1)] \quad (4.5)$$

For electric modes with frequencies

$$\omega_1 \leq \omega_k \leq \omega_s$$

we get: ($\sum_k 1 = z$)

$$N e^{\beta \hbar \omega_z} - 1 > S/\varphi + z > N e^{\beta \hbar \omega_1} - 1 \quad (4.6)$$

From Equation (4.6), we find that the N is directly proportional to S , provided that S reaches a certain threshold. In other words, any increase in total energy to the system must accompany an increase in total quanta of the system. Now, we define μ by⁵:

$$e^{-\beta \mu} = \frac{\varphi + \chi_t \Sigma (1 + n_j)}{\varphi + \chi_t \Sigma n_j e^{\beta \hbar \omega_j}} \quad (4.7)$$

$$1 - e^{-\beta \mu} = \frac{\chi_t \Sigma [\pi_j e^{\beta \hbar \omega_j} - (1 + n_j)]}{\varphi + \chi_t \Sigma n_j e^{\beta \hbar \omega_j}} \quad (4.8)$$

Substituting Equation (4.5) into Equation (4.8) yields:

$$1 - e^{-\beta \mu} = \frac{\chi_t S}{\varphi \varphi + \chi_t \Sigma n_j e^{\beta \hbar \omega_j}} \geq 0 \quad (4.9)$$

which follows that $\mu \geq 0$. Equations (4.1) and (4.2) can formally be solved by:

$$n_k = \left[1 + \frac{\varphi s_k}{\chi_t S} (1 - e^{\beta \mu}) \right] \left(\frac{1}{e^{\beta \hbar \omega_k - \beta \mu} - 1} \right) \quad (4.10)$$

Taking into the inequalities $n_k \geq 0$ and $\mu \geq 0$ to solve Equation (4.10), one finds the following relation between μ and quanta $\hbar \omega_1$, i.e.:

$$\hbar \omega_1 \geq \mu \geq 0 \quad (4.11)$$

Using Equation (4.10) and the relationship (4.11), one finds the upper limit of N in the following way:

$$N = \sum n_k < \left[1 + \frac{\varphi s_m}{\chi_t S} (1 - e^{\beta \hbar \omega_1}) \right] \sum_k \left(\frac{1}{e^{\beta \hbar \omega_k - \beta \mu} - 1} \right) \quad (4.12)$$

so that s_m is the largest among all the s_k . The condition put forth by the inequality (4.6) still holds: as μ becomes closer to $\hbar \omega_1$, n_1 takes a value that is the largest of all n_k values. As a result, the single mode, ω_1 gets strongly excited as S reaches a certain threshold.

Now, assuming that all s_k are equal to one another, the following equation holds:

$$\frac{s_k}{S} = \frac{1}{z} \quad (4.13)$$

and Equation (4.10) becomes:

$$n_k = \left[1 + \frac{\varphi}{\chi_t z} (1 - e^{\beta\mu}) \right] \left(\frac{1}{e^{\beta\hbar\omega_k - \beta\mu} - 1} \right) \quad (4.14)$$

Now, assuming that all s_k vanish except for the cases where, $k_0 = k$, the following condition holds:

$$s_k = S\delta_{k,k_0}; \quad s_{k_0} = S \quad (4.15)$$

so that Equation (4.10) now takes the form:

$$n_k = \left[1 + \frac{\varphi}{\chi_t} (1 - e^{\beta\mu}) \delta_{k,k_0} \right] \left(\frac{1}{e^{\beta\hbar\omega_k - \beta\mu} - 1} \right) \quad (4.16)$$

In the case of $\chi_t = 0$, Equation (4.10), however, takes the simple form:

$$n_k = \left(1 + \frac{s_k}{\varphi} \right) \left(\frac{1}{e^{\beta\hbar\omega_k} - 1} \right) \quad (4.17)$$

which resembles the Planck Distribution (Equation 4.3) in thermal equilibrium. However, this case also implies that $\mu = 0$ from Equation (4.7) and prevents the occurrence of Bose condensations. The nonlinear term χ_t arising from interactions with the heat bath forces the system in a distribution that would eventually lead to thermal equilibrium with the help of chemical potential energies⁶ with the main difference between Bose-Einstein condensation and Frohlich coherence is that energy in quanta are added to the system while keeping the temperature constant. It must be noted that the existence of long polarization does not require the system to homogenous. The frequencies of the electric modes discussed so far must lie within $10^9 - 10^{11} \text{Hz}$ ⁶. Furthermore, screening effect of ordered water molecules around the microtubules should not screen out oscillations in these range.

5. IONIC WAVE CONDUCTION

Molecular dynamic simulations performed by Tuszynski et al.(2005)¹ revealed that a majority of the negative charges found on the tubulin dimer is positioned on the outer surface of these dimers, with a charge ratio of 2:1 between the outer surface and the rest of the protein. This would lead us to speculate that there is a condensed ionic cloud(IC) of positive counter ions on the surface of the dimer based on Manning's theory of polyelectrolyte solutions¹⁸ that states that ionic condensation occurs along a stretch of polymer if the surface charge density exceeds a particular threshold. Based on this theory we can also assume that there is a cylindric region of depleted ions outside the IC region that acts an insulating material and shields the IC from incoming counter ions(19). As such, the ionic cloud, along with the depleted layer can act as inhomogeneous transmission lines that propagate ionic waves along the ionic cloud. In order to determine the thickness of this depleted layer, we need to take into account the Bjerrum length, which is defined as the minimum distance beyond which the thermal fluctuations between charges are stronger than the electrostatic interactions with relative permittivity, ϵ . The Bjerrum length, l_B can be found from the following equation:

$$\frac{e^2}{4\pi\epsilon_0\epsilon l_B} = k_B T \quad (5.1)$$

where $e = 1.6 \times 10^{-19} \text{J/K}$ is the electron charge, ϵ_0 is the vacuum permittivity, k_B is the Boltzmann constant and T is the physiological temperature. With $\epsilon = 80$ in the cytosol, the l_B is found to be 0.67nm (approx.). This is approximately the length of the depletion region. It is important to mention that ionic cloud condensation may only occur when the distance between charges, b is relatively shorter than l_B to justify the condition¹⁹:

$$S = \frac{l_B}{b} > 1 \quad (5.2)$$

Assuming there are about 15-18 exposed negative charges¹⁹ on the tubulin dimer (excluding the terminal tails) and the length of a dimer being 8nm, *Sataric et al.(2009)*¹⁹ estimated the linear charge spacing, $b = (0.33 - 0.44) \times 10^{-10}m$ so that S is between 16 and 21. As such, the effective charge, q_{eff} , which is found by dividing the bare net charge by S , is estimated to be $q_{eff} = (0.43 - 0.57)e/monomer$ that has been validated by experimental assays²⁰.

Furthermore, the occurrence of Manning condensation around our polymer with radius, r also depends on the salt concentration, n_{salt} , such that the following condition holds:

$$l_{Db} \geq r \quad (5.3)$$

where the Debye length, l_{Db} is defined by:

$$l_{Db}^{-1} = (8\pi n_{salt} l_B) \quad (5.4)$$

With the radius of the terminal C-tail and a tubulin dimer being 0.5nm and 2.5nm respectively, and the Debye length of the cytosol being around 10nm(21), condition (3) is satisfied for both the radii, so that we can determine the condensate thickness using the formula:

$$\lambda = A(r l_{Db})^{\frac{1}{2}}, \quad A < 1 \quad (5.5)$$

where A is a constant that depends on the Manning parameter, q_0 . With $A = 1 / 2(19)$, the respective values of λ have been found:

$$\lambda_{TD} = 2.5 \text{ nm}, \quad \lambda_{TT} = 1.1 \text{ nm} \quad (5.6)$$

At neutral pH with fewer positive counter ions, the C-terminal tails at both the monomers will form an extended conformation as the negative charges in the amino acid residues would repel each other, but at acidic pH, the negatively charged amino acid residues would be surrounded by positive counter-ions resulting in the difference of ionic condensation at different pHs. However, due to the difference in the amino acids sequence in the C-tails and their resulting pKa values, the amount of ionic condensation will vary from the α C-tails to the β C-tail. As a result, at acidic pHs, the β C-tail will be shorter than the α C-tails. *Priel et al(2005)*²². Depending on whether condition (2) holds true for our system, the β C-tail is more likely to bind to the corresponding tubulin surface. When $b > l_B$ thermal fluctuations in the system would cause both the C-tails will oscillate with a certain frequency. Due to the shortened overall length of the β C-tail, it is more likely that $b < l_B$ will occur where b now defines the minimum distance between the opposing charges in the β C-tail and the tubulin surface, so that the C-tail now binds to the surface. This is supported by transitions model for C-tails in *Priel et al.(2005)*²². However, since our system is dynamic, this binding is assumed to be temporary.

Now, we estimate the capacitance of our ionic cloud inn a single dimer(including C-tails) with the following formula

$$C_0 = \frac{2\pi\epsilon_0 e l}{\ln\left(1 + \frac{l_B}{r_{IC}}\right)} \quad (5.7)$$

where r_{IC} is the effective radius of the ionic cloud. We find that the $C_{TD} = 1 \times 10^{-16}F$ for the tubulin dimer where ($r_{IC} = r_{TD} + \lambda_{TD} = 5.0nm$) and $l = 8nm$. Similarly, $C_{TT} = 0.26 \times 10^{-16}F$ where $r_{IC} = 1.5nm$. However, l is the extended effective length, $l_{TT}^{eff} = 4.5nm - 2.5nm = 2nm$ indicating that portion of the C-tail attached to the dimer is already embedded in ionic cloud of C_{TD} . The capacitances, C_{TD} and C_{TT} respond to a parallel arrangement, so that the total capacitance of our ionic cloud in a single dimer is given by:

$$C_0 = C_{TD} + 2C_{TT} = 1.92 \times 10^{-16}F \quad (5.8)$$

To account for the difference in C_{TT} due to the difference in structure for α and β tails, we introduce the reduced factor of nonlinearity for the different values of b (23) by:

$$b_0 = \frac{b_\alpha b_\beta}{b_\alpha + b_\beta} \quad (5.9)$$

where the b_α and b_β correspond to their respective C-terminal tails.

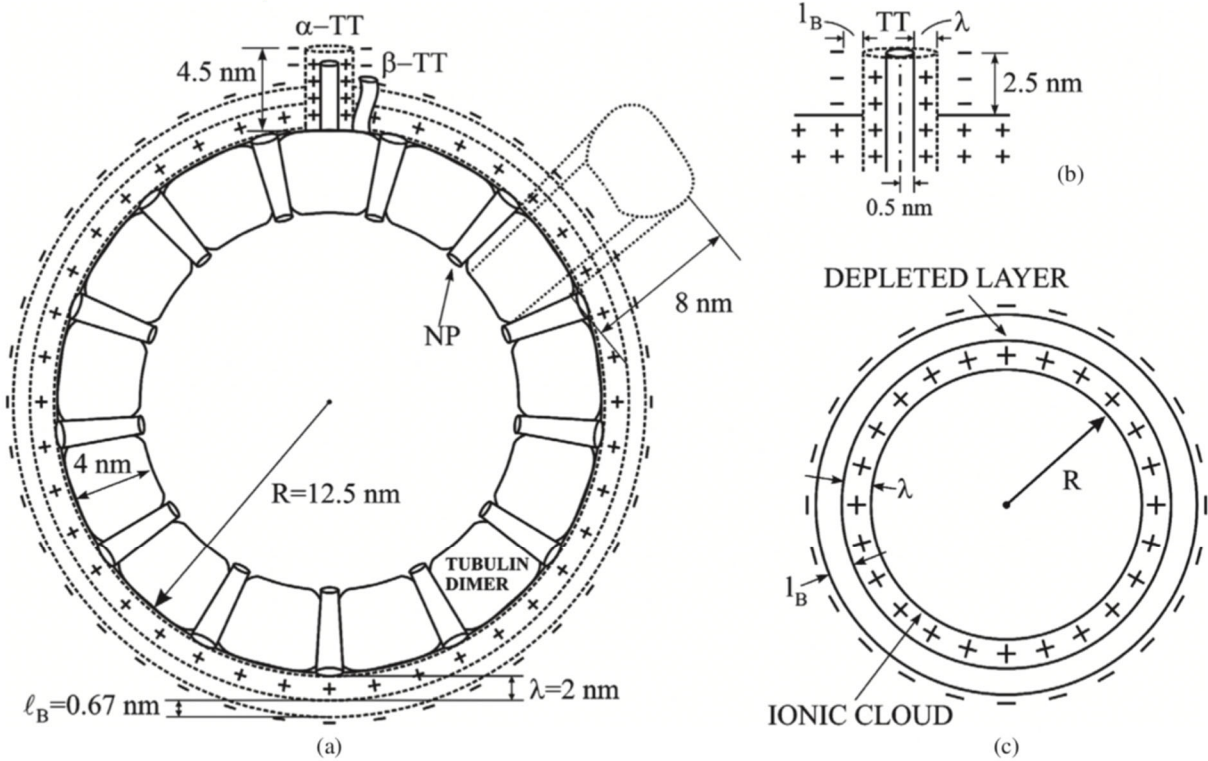


Figure 5. Geometry of MT with ionic cloud and depleted layer and their respective dimensions²⁸.

As previously mentioned, thermal fluctuations will cause the C-tails to oscillate leading to change in their l_{TT}^{eff} . This is approximated by the oscillating function:

$$\Delta l_{TT}^{eff} = l_{TT}^{eff} \sin[\Omega(t - t_0)] \cong l_{TT}^{eff} \Omega(t - t_0) \quad (5.10)$$

As such, the change in capacitance follows a linear dependence on the l_{TT}^{eff} , so that the change in capacitance is accounted by:

$$\Delta C_{TT} = C_0 \Gamma_0 \Omega(t - t_0) \quad (5.11)$$

In both Equations (5.10) and (5.11), the frequency Ω is less than the inverse charging time of our capacitor due to the viscous force in the cytosol and Γ_0 is a dimensionless parameter (its significance is discussed later in this section). Using the aspects of TTs dynamics described in equations (5.9) and (5.11), the charge, Q of a dimer can be expressed as²³:

$$Q = C_0 [1 - \Gamma_0 \Omega(t - t_0) - b_0 v] v \quad (5.12)$$

where Equation (5.12) accounts the change in total charge affected by the oscillations of the C-tails in the second term and the difference in structure of the two tails in the third term.

Furthermore, the resistance for our system has been estimated at $R_0 = 6 \cdot 2 \times 10^7 \Omega$ (23) for all 13 protofilaments. Taking the reported measured value of MT ionic conductivity of $\sigma = (0.15 \pm 0.01) \text{ Sm}^{-1}$ the conductance, G_0 has been estimated at

$$G_0 = G_1 + G_2 = (0.293 + 0.780) \text{ nS} = 0.107 \text{ nS}$$

where the conductance for the two NPs differ because of the difference in their sizes due to their relative positions in adjacent dimers, which are situated on adjacent dimers

Based on all the parameters we have estimated so far, we can model the microtubule as transmission lines for ionic waves that mainly travel along the axis of the MTs as longitudinal waves that are affected by the terminal tails as well as conductance through the nanopores. Therefore, each dimer is assumed to have a current, I_n , where n indicates the n th dimer of the MT. and voltage across a dimer is defined as V_n . The nonlinear capacitance, C_0 with the charge Q_n of the n th dimer corresponding to the n th site of the ladder in figure 5 is in parallel with the total conductance, G_0 of the two terminal tails.

Using Kirchhoff's current and voltage laws, we can define the relationship between the adjacent currents I_n, I_{n+1} with resistance R_0 , as:

$$I_n - I_{n+1} = \frac{\delta Q_n}{\delta t} + G_0 V_n \quad (5.13)$$

where,

$$\frac{\delta Q_n}{\delta t} = C_0 \frac{\delta V_n}{\delta t} - C_0 \Gamma_0 \Omega V_n - C_0 \Gamma_0 \Omega (t - t_0) \frac{\delta V_n}{\delta t} - 2b_0 C_0 V_n \frac{\delta V_n}{\delta t} \quad (5.14)$$

and the relationship between the voltages, V_{n-1}, V_n with resistance, R_0 as:

$$V_{n-1} - V_n = R_0 I_n \quad (5.15)$$

Next, we use the auxiliary function, $U(x, t)$ to unify Equations (5.13) and (5.15) into one equation. The function is defined by:

$$U_n = Z^{1/2} I_n = Z^{-1/2} V_n \quad (5.16)$$

where Z is the characteristic impedance of our dimer and is defined as:

$$Z = \frac{1}{\omega C_0} \quad (5.17)$$

Using the transformations from Equation (5.16) and using Taylor series to expand U_{n+1} and U_{n-1} , Equation (5.13) becomes:

$$\frac{\delta U}{\delta x} + \frac{l}{2} \frac{\delta^2 U}{\delta x^2} + \frac{l^2}{6} \frac{\delta^3 U}{\delta x^3} + \frac{Z C_0}{l} \left[\frac{\delta U}{\delta t} - \Gamma_0 \Omega U - \Gamma_0 \Omega (t - t_0) \frac{\delta U}{\delta t} - Z^{1/2} b_0 U \frac{\delta U}{\delta t} \right] - \frac{Z G_0 U}{l} = 0 \quad (5.18)$$

and Equation (5.15) becomes:

$$-\frac{\delta U}{\delta x} + \frac{l}{2} \frac{\delta^2 U}{\delta x^2} - \frac{l^2}{6} \frac{\delta^3 U}{\delta x^3} - \frac{U R_0}{Z l} = 0 \quad (5.19)$$

Adding Equations (5.18) and (5.19), we get the following travelling form of our auxiliary function, $U(x, t)$:

$$l \frac{\delta^2 U}{\delta x^2} + \frac{Z C_0}{l} \left[\frac{\delta U}{\delta t} - \Gamma_0 \Omega U - \Gamma_0 \Omega (t - t_0) \frac{\delta U}{\delta t} - Z^{1/2} b_0 U \frac{\delta U}{\delta t} \right] - \frac{U R_0}{Z l} - Z G_0 U = 0 \quad (5.20)$$

However, now we must nondimensionalize our auxiliary function, $U(x, t)$ with dimensionless space and time variables (ξ, τ) to $U(\xi, \tau)$ using the following substitutions:

$$U(x, t) = U\left(\frac{x}{l} - \frac{v v_0 t}{v_0 l}\right) = U(\xi) \quad (5.21)$$

where,

$$\xi = \frac{x}{l} - \tau, \quad v_0 = \frac{l}{T_0}, \quad s = \frac{v}{v_0}, \quad \tau = s \frac{t}{T_0} \quad (5.22)$$

and with the following transformations:

$$\begin{aligned}\frac{\delta U}{\delta x} &= \frac{1}{l} \frac{\delta U}{\delta \xi}; \quad \frac{\delta^3 U}{\delta x^3} = \frac{1}{l^3} \frac{\delta^3 U}{\delta \xi^3} \\ \frac{\delta U}{\delta t} &= \frac{s}{T_0} \frac{\delta U}{\delta \tau} = -\frac{s}{T_0} \frac{\delta U}{\delta \xi}\end{aligned}\quad (5.23)$$

Equation (5.20) becomes:

$$\begin{aligned}\left(ZC_0 \frac{s}{T_0} - 2\right) \frac{\delta U}{\delta \tau} + \frac{1}{3} \frac{\delta^3 U}{\delta \xi^3} + ZC_0 \Gamma_0 \Omega (\xi - \xi_0) \frac{\delta U}{\delta \xi} + Z^{\frac{3}{2}} b_0 C_0 U \frac{s}{T_0} \frac{\delta U}{\delta \xi} \\ + (ZG_0 + Z^{-1}R_0 - ZC_0 \Gamma_0 \Omega) U = 0\end{aligned}\quad (5.24)$$

From Equation (5.24) we deduce the down-limit of the ionic wave speed as:

$$ZC_0 \frac{s}{T_0} - 2 > 0$$

Now we make Equation (5.24) with the following substitutions:

$$\frac{\delta U}{\delta \tau} + \beta \delta \frac{3U}{\delta \xi^3} + \gamma(\xi) \frac{\delta U}{\delta \xi} + \alpha U \frac{\delta U}{\delta \xi} + \partial U = 0 \quad (5.25)$$

where,

$$\begin{aligned}\alpha &= \frac{2Z^{3/2}b_0C_0s}{T_0 \left(\frac{ZC_0s}{T_0} - 2\right)}; \quad \beta = \frac{1}{3 \left(\frac{ZC_0s}{T_0} - 2\right)}; \quad \partial = \frac{ZG_0 + Z^{-1}R_0 - ZC_0\Gamma_0\Omega}{\left(\frac{ZC_0s}{T_0} - 2\right)} \\ \gamma_0 &= \frac{ZC_0\Gamma_0\Omega}{\left(\frac{ZC_0s}{T_0} - 2\right)}; \quad \gamma = \gamma_0(\xi - \xi_0)\end{aligned}\quad (5.26)$$

Using the following transformations on the space-time variables, (ξ, τ) to (ρ, θ) :

$$\rho(\xi, \tau) = (\xi - \xi_0)e^{-\gamma_0\tau}; \quad \theta(\tau) = \frac{1}{-3\gamma_0} e^{-3\gamma_0\tau}; \quad U(\xi, \tau) = W(\rho, \theta)e^{-2\gamma_0\tau} \quad (5.27)$$

Equation (5.25) becomes:

$$\frac{\delta W}{\delta \theta} + \alpha W \frac{\delta W}{\delta \rho} + \beta \frac{\delta^3 W}{\delta \rho^3} + \frac{\kappa}{\theta} W = 0 \quad (5.28)$$

where κ is the new parameter:

$$\kappa = \frac{2}{3} - \frac{\partial}{3\gamma_0} \quad (5.29)$$

which reduces Equation (5.25) from 5 terms to 4 terms. Therefore, Equation (5.28) has similar form of a Davydov's soliton when assumes that the last term vanishes under the “thermal ratchet” action of TTs. This thermal ratcheting action is similar to the stretching of hydrogen bonds to capture energy in the form a soliton as given by the nonlinear term $\alpha W \frac{\delta W}{\delta \rho}$ in Equation (5.28).

Furthermore, Equation (5.29) gives the following condition²³:

$$ZG_0 + \frac{R_0}{Z} = 3ZC_0\Gamma_0\Omega; \Rightarrow \Gamma_0 = \frac{G_0}{3C_0\omega} + \frac{C_0R_0\omega}{3} \approx 0.22 \quad (5.30)$$

given that $\Omega = \omega = 2.78 \times 10^7 s^{-1}$ and $G_0 = 1.1 \times 10^{-9} S$. Then, the spatial inhomogeneity parameter is determined by $\Gamma_0\Omega = 6.12 \times 10^6 s^{-1}$. Since the inverse characteristic discharge time, $T_0^{-1} = 8.33 \times 10^7 s^{-1}$ and $\Gamma_0\Omega < T_0^{-1}$, the $\Gamma_0\Omega$ parameter accounts for the viscous effect of the aqueous solution surrounding the TTs in the cytosol.

Eventually, Equation (5.28) has the following solution with $\kappa = 0$ ²³:

$$W(\xi, \tau) = \frac{U_0}{\cosh^2\left\{\left[\frac{\alpha U_0}{4\beta} e^{-2\gamma_0 \tau}\right]^{1/2} [\xi - \xi_0(1 - e^{\gamma_0 \tau}) + \frac{\alpha U_0}{3\beta}(1 - e^{3\gamma_0 \tau} e^{-2\gamma_0 \tau})]\right\}} \quad (5.31)$$

so that the function, $U(\xi, \tau)$ has the form:

$$U(\xi, \tau) = \frac{U_0 e^{-2\gamma_0 \tau}}{\cosh^2\left\{\left[\frac{\alpha U_0}{4\beta} e^{-2\gamma_0 \tau}\right]^{1/2} [\xi - \xi_0(1 - e^{\gamma_0 \tau}) + \frac{\alpha U_0}{3\beta}(1 - e^{3\gamma_0 \tau} e^{-2\gamma_0 \tau})]\right\}} \quad (5.32)$$

Considering all the parameters that have been estimated so far ($C_0, R_0, G_0, Z, \Gamma_0, \Omega$), the dimensionless parameters from Equation (5.26) lie between 0 and 1 ($\beta = 0.34$, $\partial = 0.49$, $\gamma_0 = 0.17$). Performing a non-linear study with these parameters and the nonlinearity α , Sekulic et al (2011)²³ determined that when these parameters are balanced with respect to each other, the soliton solution in Equation (5.32) disperses over certain length of the microtubule polymer as the amplitude of the soliton decays over length and time. On the other hand, when the nonlinearity, α is significantly higher than these dimensionless parameters, the soliton maintains its amplitude over length and time.

Assuming that ICS behaves like a particle in a viscous liquid (in our case, the cytosol), the ICS will be subjected to deceleration due to the viscous force in the cytosol, F_{vis} . As a result, the speed of the ICS, v will follow an exponential decay towards the down-limit, $ZC_0 \frac{s}{T_0} = 2$. However, since MTs have an intrinsic electric field²⁸, the ICS will also experience acceleration due to the force of the electric field F_{if} . As a result, we have to consider two force parameters acting on the flow of our ICS:

$$F_{vis} = -kv. \quad (5.33)$$

and

$$F_{if} = Q \cdot E_i \quad (5.34)$$

where, k depends on the dynamical viscosity of the cytosol, and E_i is the strength of the electric field that depends on the pH conditions and length of the MTs²⁸.

6. INDUCTANCE IN THE C TERMINAL TAILS

In the model we have developed so far, we have considered the flow of ionic cloud (counter-ions) along the axis of the MT protofilament. However, there is also leakage of counter-ions via the nanopores. In other words, a current travels in the direction normal to the protofilament axis. In order to extend our model to the C-terminal tails, we introduce the possibility of a magnetic flux accompanying this current flow. The magnetic flux, Φ , for such a current, I_{TT} is given by:

$$\Phi = \frac{\mu_0}{l_{TT}^{eff}} \pi R_{TT}^2 I_{TT} \cos \theta; \quad 0 \leq \theta \leq \frac{\pi}{2} \quad (6.1)$$

and the induced voltage, V_F that results from Faraday's law of induction is given,

$$V_F = -L \frac{dI_{TT}}{dt} \quad (6.2)$$

so that the inductance of the C-tails, L is given by:

$$L = \frac{\mu_0}{l_{TT}^{eff}} \pi R_{TT}^2 \cos \theta \quad (6.3)$$

where $R_{TT} = \lambda_{TT} + 0.5nm = 1.1nm$ from Equation (5.6) and the magnetic permeability²⁶, $\mu_0 = 4\pi \times 10^{-7}H/m$. Since the C-terminal tails extend over the nanopores, we assume that the current travels along the axis of the tails and through the nanopores via conduction. Transition simulation models performed by *Priel et al. (2005)*²³ determined that the C-tails in both the α and β -monomers exist in combinations where either both the C-tails are erect or where the α C-tail is erect and the β C-tail is bent and bound to the tubulin surface. Now, according to Fleming's right hand grip rule the resulting magnetic field would have a counter-clockwise flow if the ionic flow towards the tip of the C-tail or clockwise magnetic field would result if this ionic cloud leaks into the nanopores. Since the C-tail is the most flexible part of the tubulin monomer, we assume that there is an electromotive force exerted on the tail according to Fleming's right hand rule. Therefore, if the ionic flow is towards the nanopores on the tubulin surface, then the corresponding C-tail will be shortened because of the force. On the other hand, then an ionic flow out of the nanopore, would cause the C-tail to extend.

There is a similar effect on the length of the C-tail when the pH of the cytosol changes. Because the C-tail consists largely negative anion residues that repel one another, the C-tail will be fully extended in the absence of positive counter-ions. This will be the case when $pH \geq 7$, but at $pH \leq 7$, the number of positive counter-ions increases that neutralize the negative charges and cause the C-tail to shorten.

This electromotive force initiates the change in the length of the C-tail accompanied by the flow of ionic cloud into or out of the nanopores. This change in flow would change the pH in the vicinity of the C-tail that maintains the length of the C-tail.

Since the ionic flow can leak in both directions of the nanopores, we propose that there is an alternating current present along the axis of the C-tails and the nanopores normal to the protofilament axis.

DISCUSSION

So far, we have developed our model to include the conduction pathway along the alpha-helix on the tubulin surface as well as the ionic wave conduction that travels mainly along the protofilament axis. Energy propagates from the hydrolysis of the GTP to GDP in the form of excited electrons. These excited electrons then cause a change in the concentration of positive counter-ions in the ionic cloud. For obvious reasons, we assume that the electronic conduction moves in parallel with the ionic wave propagation, with the ionic wave propagation lagging slightly behind the electronic conduction. Therefore, the ionic flow that eventually leaks into or out of the nanopores is accompanied by the electric conduction in the tubulin alpha helix.

The electric conduction along the alpha helix is maintained because of the existence of Davydov's solitons and is propagated to the π bonds within aromatic residues since the distances exist within the limits of the Foerster distance. Now, communication between different energy eigenstates in the peptide bonds is achieved by Frohlich's coherence. This is the part where Frohlich's cancer problem hypothesis⁶ might be of significance. The Warburg's effect states that cancer cells undergo transition from aerobic glycolysis to anaerobic glycolysis. Since anaerobic glycolysis is more readily available (less efficient than aerobic glycolysis), we can presume that the energy available to the tumor tissues will vary in magnitude due to the heterogeneity of most late stage cancers that this model aims to explain. Since microtubules mainly form the cytoskeleton and have variety of roles in the traffic of the cell signalling pathways as well as transport of proteins, we propose that the difference in energy would significantly affect this heterogeneity in the cell.

Based on the mathematical model that translates the energy propagation from the hydrolysis of GTP to GDP to the development of an alternating current along the axis of the C-tails and the nanopores by virtue of their nonlinear terms, we aim to further the investigation by implementing this model into a series of deep neural networks composed of Hopfield nets made of qubits and classical neural nets made of classical bits. Depending on whether the C-tail is in bent or erected form, would determine whether proteins kinesin/dynein or MAP2 bind to the tubulin surface, which will be the output for the neural networks and hence determine the heterogeneity in the cytoskeletal network as well as the energy propagation.

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